

Inverse Trig – Curve Sketching

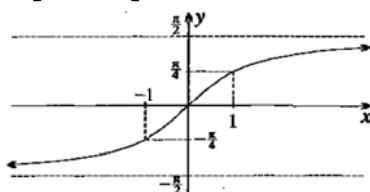
- Sketch the function, stating the domain and range and whether it is even, odd or neither:
 - $y = \tan^{-1} x$
 - $y = \cos^{-1} x$
 - $y = \sin^{-1} x$
- Sketch each function, using appropriate translations of $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$. State the domain and range, and whether it is even, odd or neither.
 - $y = \sin^{-1}(x - 1)$
 - $y = \cos^{-1}(x + 1)$
 - $y - \frac{\pi}{2} = \tan^{-1} x$
- Sketch each function by stretching $y = \sin^{-1} x$, $y = \cos^{-1} x$ and $y = \tan^{-1} x$ horizontally or vertically as appropriate. State the domain and range, and whether it is even, odd or neither.
 - $y = 2 \sin^{-1} x$
 - $y = \cos^{-1} 2x$
 - $y = \frac{1}{2} \tan^{-1} x$
- Sketch each function by reflecting in the x - or y -axis as appropriate. State the domain and range, and whether it is even, odd or neither.
 - $y = -\cos^{-1} x$
 - $y = \tan^{-1}(-x)$
 - $y = -\sin^{-1}(-x)$
- Sketch each function. Stating the domain and range, and whether it is even, odd or neither:
 - $y = 3 \sin^{-1} 2x$
 - $y = \frac{1}{2} \cos^{-1} 3x$
 - $y = \tan^{-1}(x - 1) - \frac{\pi}{2}$
 - $3y = 2 \sin^{-1} \frac{\pi}{2}$
 - $\frac{1}{2}y = 2 \cos^{-1}(x - 2)$
 - $y = \frac{1}{4} \cos^{-1}(-x)$
- a. Consider the function $y = 4 \sin^{-1}(2x + 1)$.
 - Solve $-1 \leq 2x + 1 \leq 1$ to find the domain.
 - Solve $-\frac{\pi}{2} \leq \frac{y}{4} \leq \frac{\pi}{2}$ to find the range.
 - Hence sketch the graph of the function.
- Use similar steps to find the domain and range of each function, and hence sketch it:
 - $y = 3 \cos^{-1}(2x - 1)$
 - $y = \frac{1}{2} \sin^{-1}(3x + 2)$
 - $y = 2 \tan^{-1}(4x - 1)$
- a. i. Sketch the graph of $y = \cos^{-1} x$ and $y = \sin^{-1} x - \frac{\pi}{2}$ on the same set of axes.
 - Hence show graphically that $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}$.
- Use a graphical approach to show that:
 - $\tan^{-1}(-x) = -\tan^{-1} x$
 - $\cos^{-1} x + \cos^{-1}(-x) = \pi$
- a. Determine the domain and range of $y = \sin^{-1}(1 - x)$
 - Complete the following table. Hence sketch the graph of the function.

x	0	1	2
y			
 - About which line are the graphs of $y = \sin^{-1}(1 - x)$ and $y = \sin^{-1} x$ symmetrical?
- Find the domain and range, draw up a table of values if necessary, and then sketch:
 - $y = 2 \cos^{-1}(1 - x)$
 - $y = \tan^{-1}(\sqrt{3} - x)$
 - $y = \frac{1}{3} \sin^{-1}(2 - 3x)$
- Sketch the graph of each function using the methods of this section:
 - $-y = \sin^{-1}(x + 1)$
 - $y = -\tan^{-1}(1 - x)$
 - $y + \frac{\pi}{2} = \frac{1}{2} \cos^{-1}(-x)$
- Sketch these graphs, stating whether each function is even, odd or neither:
 - $y = \sin(\sin^{-1} 2x)$
 - $y = \cos\left(\cos^{-1} \frac{\pi}{2}\right)$
 - $y = \tan(\tan^{-1}(x - 1))$
- Sketch graphs of each of the following, stating their domain and range.
 - $y = 2 \sin^{-1} x$
 - $y = 2 \sin^{-1} 3x$
 - $y = 3 \sin^{-1}(2x + 1)$
 - $y = 3 \sin^{-1} 2x + 1$
 - $y = 2 \tan^{-1} x$
 - $y = 2 \tan^{-1}(x - 1)$
 - $y = 2 \cos^{-1}\left(\frac{x}{2}\right)$
 - $y = 2 \sin^{-1}(-x)$
 - $y = \sin(\cos^{-1} x)$, by first simplifying $\sin(\cos^{-1} x)$

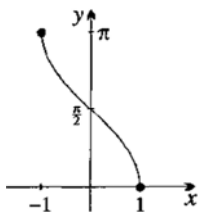
13. Consider the functions $f(x) = 2 \sin^{-1}(2x - 1)$ and $g(x) = 2 \sin^{-1}(1 - 2x)$
- Find the domain and range of each function.
 - Show that $g(x) = -f(x)$.
 - Sketch the graphs of the two functions.
14. If $f(x) = 3 \cos^{-1}\left(\frac{x}{2}\right)$, determine the inverse function f^{-1} and specify the domain and range of f^{-1} .
15. Consider the function $f(x) = \sin 2x$
- Sketch the graph of $f(x)$, for $-\pi \leq x \leq \pi$.
 - What is the largest domain containing $x = 0$ for which $f(x)$ has an inverse function?
 - Sketch the graph of $f^{-1}(x)$ by reflecting in the line $y = x$.
 - Find the equation of $f^{-1}(x)$, and state its symmetry.
16. a. What is the domain of $y = \sin \cos^{-1} x$? Is it even, odd or neither?
- By considering the range of $\cos^{-1} x$, explain why $\sin \cos^{-1} x \geq 0$, for all x in its domain.
 - By squaring both sides of $y = \sin \cos^{-1} x$ and using the identity $\sin^2 \theta + \cos^2 \theta = 1$, show that $y = \sqrt{1 - x^2}$.
 - Hence sketch $y = \sin \cos^{-1} x$
 - Use similar methods to sketch the graph of $y = \cos \sin^{-1} x$.
17. Consider the function $y = \tan^{-1} \tan x$
- State its domain and range, and whether it is even, odd or neither.
 - Simplify $\tan^{-1} \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.
 - What is the period of the function?
 - Use the above information and a table of values if necessary, to sketch the function.
18. In a worked exercise, $y = \cos^{-1} \cos x$ is sketched. Use $\sin^{-1} t = \frac{\pi}{2} - \cos^{-1} t$ and simple transformations to sketch $y = \sin^{-1} \cos x$. State its symmetry.

Answer

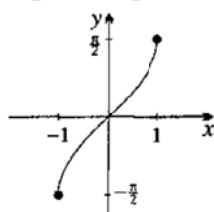
1. a. domain: all real x , range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd;



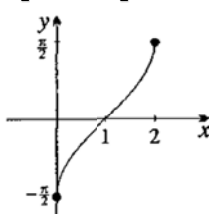
- b. domain: $-1 \leq x \leq 1$, range: $0 \leq y \leq \pi$, neither;



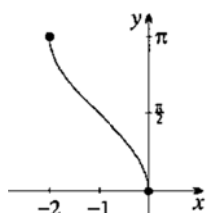
- c. domain: $-1 \leq x \leq 1$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd



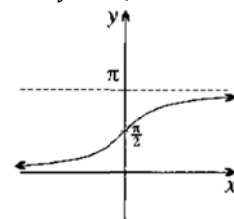
2. a. domain: $0 \leq x \leq 2$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, neither;



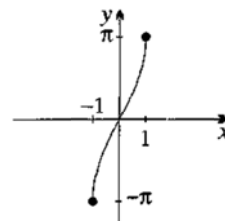
- b. domain: $-2 \leq x \leq 0$, range: $0 \leq y \leq \pi$, neither;



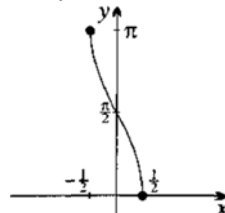
- c. domain: all real x , range: $0 < y < \pi$, neither.



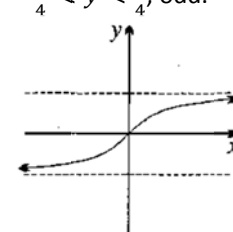
3. a. domain: $-1 \leq x \leq 1$, range: $-\pi \leq y \leq \pi$, odd;



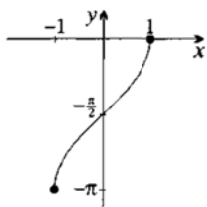
- b. domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$, range: $0 \leq y \leq \pi$, neither;



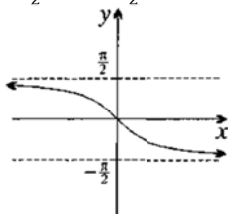
- c. domain: all real x , range: $-\frac{\pi}{4} < y < \frac{\pi}{4}$, odd.



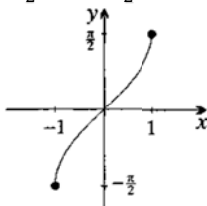
4. a. domain: $-1 \leq x \leq 1$, range: $-\pi \leq y \leq 0$, neither;



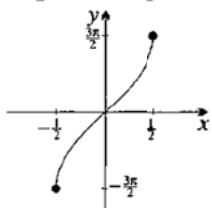
- b. domain: all real x , range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd;



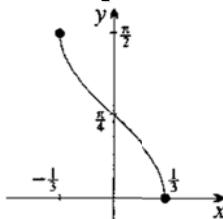
- c. domain: $-1 \leq x \leq 1$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, odd



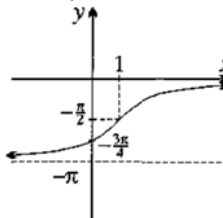
5. a. domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$, range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$, odd;



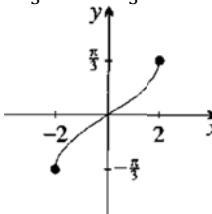
- b. domain: $\frac{1}{3} \leq x \leq \frac{1}{3}$, range: $0 \leq y \leq \frac{\pi}{2}$, neither;



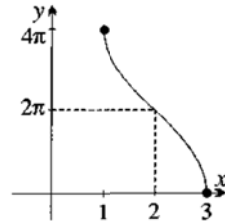
- c. domain: all real x , range: $-\pi < y < 0$, neither;



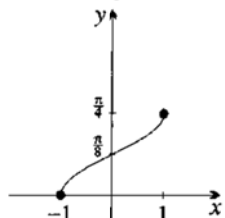
- d. domain: $-2 \leq x \leq 2$, range: $-\frac{\pi}{3} \leq y \leq \frac{\pi}{3}$, odd;



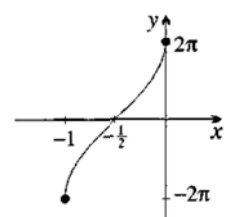
- e. domain: $1 \leq x \leq 3$, range: $0 \leq y \leq 4\pi$, neither;



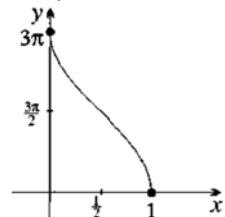
- f. domain: $-1 \leq x \leq 1$, range: $0 \leq y \leq \frac{\pi}{4}$, neither.



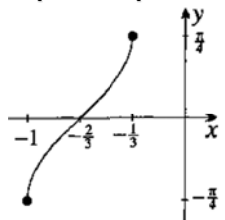
6. a. i. $-1 \leq x \leq 0$, ii. $-2\pi \leq y \leq 2\pi$;



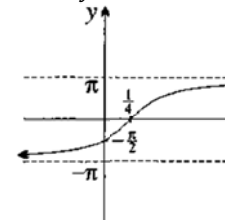
- b. i. domain: $0 \leq x \leq 1$, range: $0 \leq y \leq 3\pi$,



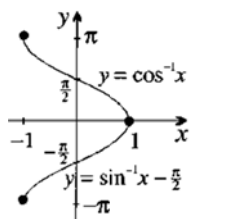
- ii. Domain: $-1 \leq x \leq -\frac{1}{3}$, range: $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$,



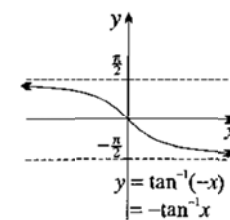
- iii. Domain: all real x , range: $-\pi < y < \pi$



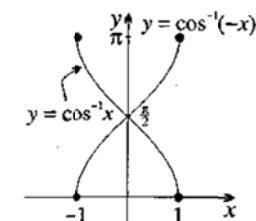
7. a. i.



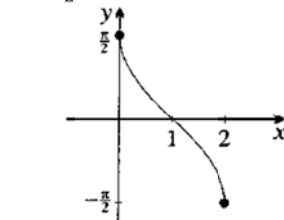
- b. i.



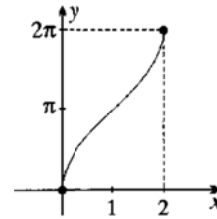
- ii.



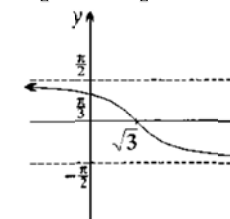
8. a. domain: $0 \leq x \leq 2$, range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$; b. $y = \frac{\pi}{2}, 0, -\frac{\pi}{2}$; c. $x = \frac{1}{2}$



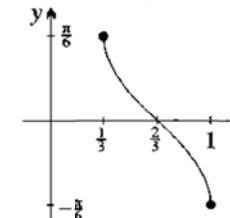
9. a. domain: $0 \leq x \leq 2$, range: $0 \leq y \leq 2\pi$;



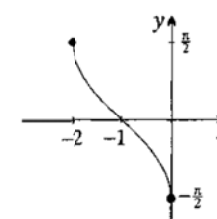
- b. domain: all real x , range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$;



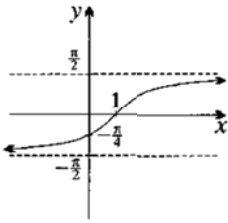
- c. domain: $\frac{1}{3} \leq x \leq 1$, range: $-\frac{\pi}{6} \leq y \leq \frac{\pi}{6}$



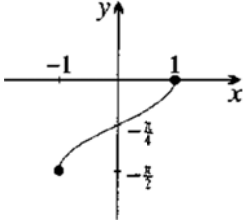
10. a.



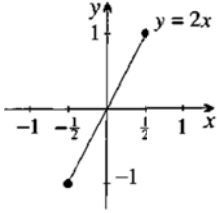
b.



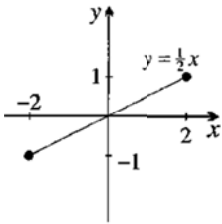
c.



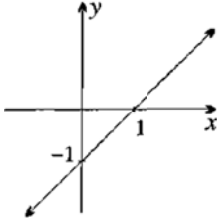
11. a. odd;



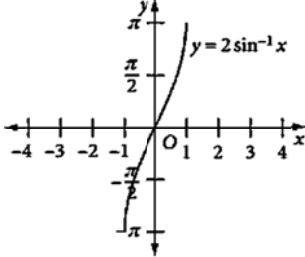
b. odd;



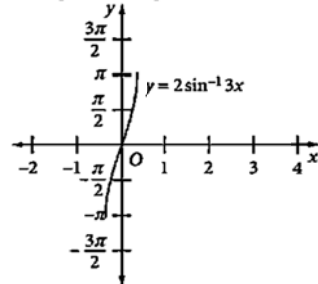
c. neither



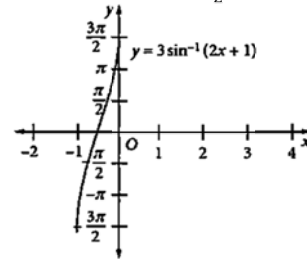
12. a. $-1 \leq x \leq 1$, $-\pi \leq y \leq \pi$;



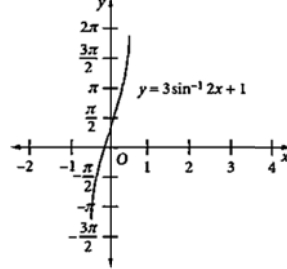
b. $-\frac{1}{3} \leq x \leq \frac{1}{3}$, $-\pi \leq y \leq \pi$;



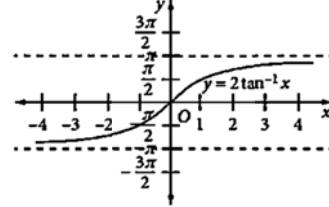
c. $-1 \leq x \leq 0$, $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$;



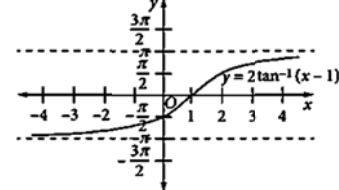
d. $-\frac{1}{2} \leq x \leq \frac{1}{2}$, $-\frac{3\pi}{2} + 1 \leq y \leq \frac{3\pi}{2} + 1$;



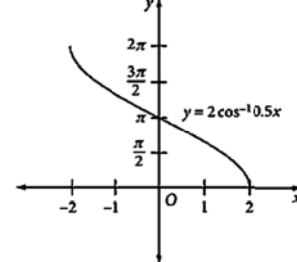
e. all real x , $-\pi \leq y \leq \pi$;



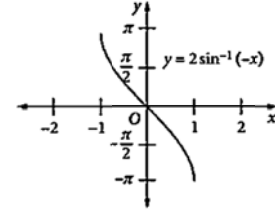
f. all real x , $-\pi \leq y \leq \pi$;



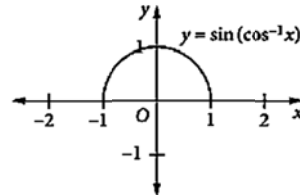
g. $-2 \leq x \leq 2$, $0 \leq y \leq 2\pi$;



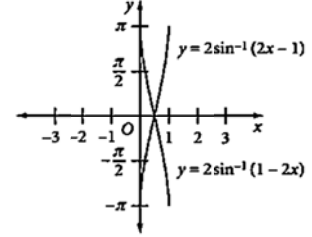
h. $-1 \leq x \leq 1$, $-\pi \leq y \leq \pi$;



i. $-1 \leq x \leq 1$, $0 \leq y \leq 1$

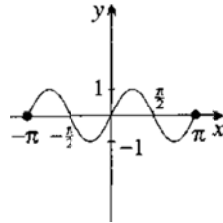


13. a. Both have domain $0 \leq x \leq 1$, range $-\pi \leq y \leq \pi$; c. see below



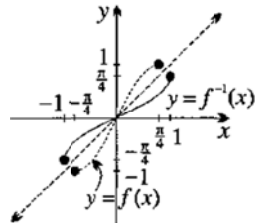
14. $y = 2 \cos \frac{x}{3}$, $0 \leq x \leq 3\pi$, $-2 \leq y \leq 2$

15. a.



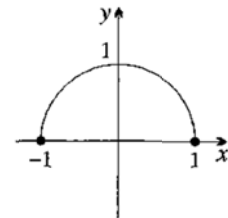
b. $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

c.

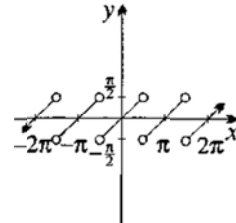


d. $f^{-1}(x) = \frac{1}{2} \sin^{-1} x$, odd

16. a. $-1 \leq x \leq 1$, even; b. $0 \leq \cos^{-1} x \leq \pi$, so $\sin \cos^{-1} x \geq 0$; d. see diagram below



17. a. domain: all real x , $x \neq \frac{(2n+1)\pi}{2}$, where n is an integer, range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$, odd; b. x ; c. π ; d. see diagram below



18. $\sin^{-1} \cos x = \frac{\pi}{2} - \cos^{-1} \cos x$, so we reflect in the x -axis and then shift $\frac{\pi}{2}$ units up. It's even.

